

Problems	Loss	Reference
1. Mach-Zehnder Modulator (MZM)	10 to 30 dB/cm (for Depleted PN-junction)	Rahim A, Hermans A, Wohlfeil B, Petousi D, Kuyken B, Van Thourhout D, Baets R. Taking silicon photonics modulators to a higher performance level: state-of-the-art and a review of new technologies. Advanced Photonics. 2021 Mar 1;3(2):024003-.
2. Ring Resonator Add-Drop Filter	1.3 dB (drop port)	Mirza A, Gloekler RE, Thompson J, Pasricha S, Nikdsast M. Experimental analysis of adiabatic silicon photonic microring resonators under process variations. IEEE Photonics Technology Letters. 2024 Jan 29;36(6):417-20.
3. Dual-Ring Resonator Tunable Filter	3 dB	
4. 4-Channel WDM using Cascaded MZIs	0.45 dB	Yi Q, Zheng S, Yan Z, Cheng G, Xu F, Li Q, Shen L. Silicon photonic flat-top WDM (de) multiplexer based on cascaded Mach-Zehnder interferometers for the 2 μm wavelength band. Optics Express. 2022 Jul 15;30(15):28232-41.
5. Tunable 1×4 Optical Switch	5.3 dB	Ghanaatian Z, Shafiee A, Nikdast M. Enhanced silicon photonic switch fabrics with variation-aware optimized Mach-Zehnder interferometers. InSilicon Photonics XX 2025 Mar 20 (Vol. 13371, pp. 45-51). SPIE.
6. Balanced Coherent Receiver	6.5 dB	Sun W, Cao H, Xie J, Xiang Y, Chen T, Guo J, Li H, Yu Z, Dai D. Ultralow-LO-Power Silicon Photonic Coherent Receiver with Balanced Ge/Si Avalanche Photodiodes. ACS Photonics. 2025 Mar 11;12(4):1793-801.
7. 16-Channel AWG	2.2 dB	Shen X, Li C, Zhao W, Li H, Shi Y, Dai D. Ultra-Low-Crosstalk Silicon Arrayed-Waveguide Grating (De) multiplexer with 1.6-nm Channel Spacing. Laser & Photonics Reviews. 2024 Jan;18(1):2300617.
8. 2×2 Thermo-Optic Switch	0.7 dB	Zhang L, Yang T, Zhang E, Zhang S, Yang S, Yang X. Single-stage 2× 2 silicon thermo-optic switch with crosstalk<− 35 dB over 40 nm and Pπ< 8 mW.
9. Y-Branch Power Splitter	0.28 dB	Zhang Y, Yang S, Lim AE, Lo GQ, Galland C, Baehr-Jones T, Hochberg M. A compact and low loss Y-junction for submicron silicon waveguide. Optics express. 2013 Jan 11;21(1):1310-6.
10. 90-Degree Optical Hybrid	1.1 dB	Chen Y, Xie X, Sun Y, Pan W, Yan L. Compact optical 90° hybrid based on a wedge-shaped 2× 4 MMI coupler and a 2× 2 MMI coupler on a thin-film lithium niobate platform. Optics Letters. 2024 Feb 20;49(5):1145-8.
11. Add-Drop Multiplexer (OADM)	The same as Add-Drop MRR Filter.	
12. Tunable Ring Resonator Bank (4 Rings)	0.08 dB	Sunny F, Shafiee A, Charbonnier B, Nikdast M, Pasricha S. Comet: a cross-layer optimized optical phase-change main memory architecture. In2024 Design, Automation & Test in Europe Conference & Exhibition (DATE) 2024 Mar 25 (pp. 1-6). IEEE.
13. 16-QAM Modulator	The same as Add-Drop MRR Filter.	
14. Wavelength Selective Switch (1×8)	5.56 dB	Cheng J, Zhang D, Zeng G, Zhou M, Ding Y, Yin Y. Low-loss and broadband 1× 8 thermo-optic selective switch using bent directional couplers. Results in Optics. 2025 Sep 12:100891.
15. Grating Coupler Array (8 Channels)	Loss per GC : 8 dB	Almeida D, Lourenço P, Fantoni A, Costa J, Vieira M. Grating coupler design for low-cost fabrication in amorphous silicon photonic integrated circuits. InPhotonics 2024 Aug 23 (Vol. 11, No. 9, p. 783). MDPI.
16. Integrated Balanced Photodetector	N/A	
17. Microring Modulator with Feedback	The same as Add-Drop MRR Filter.	

Problems	Loss	Reference
18. 2×2 Switch (PCM-based)	2 dB	Chen R, Fang Z, Froch JE, Xu P, Zheng J, Majumdar A. Broadband nonvolatile electrically controlled programmable units in silicon photonics. ACS Photonics. 2022 May 6;9(6):2142-50.
19. 16 × 16 Switch Based on Dual-Ring-Assisted MZI	10.6 dB	Guo Z, Lu L, Zhou L, Shen L, Chen J. 16× 16 silicon optical switch based on dual-ring-assisted Mach–Zehnder interferometers. Journal of Lightwave Technology. 2017 Sep 13;36(2):225-32.
20. optical delay line	0.08 dB/m	Lee H, Chen T, Li J, Painter O, Vahala KJ. Ultra-low-loss optical delay line on a silicon chip. Nature communications. 2012 Jan;3(1):867.

Components	Loss	Reference
Multimode Interference coupler (MMI) 1X2	0.3 dB	Zhu Z, Zhong Y, Lin D, Tong X, Zhong Q, He J. Low-loss and broadband arbitrary ratios 1× 2 power splitter based on asymmetrically tapered multimode interference. Journal of Optics. 2024 May 21;26(7):075801.
MMI 1X4	0.58 dB	Hong S, Yoon J, Kim J, Kim JY, Neseli B, Yoon H, Park HH, Kurt H. Design of MMI-based 1x4 power splitters with optimized parabolic input and output ports on SOI platform. InSilicon Photonics XVIII 2023 Mar 13 (Vol. 12426, pp. 30-35). SPIE.
Y-branch	0.28 dB	Zhang Y, Yang S, Lim AE, Lo GQ, Galland C, Baehr-Jones T, Hochberg M. A compact and low loss Y-junction for submicron silicon waveguide. Optics express. 2013 Jan 11;21(1):1310-6.
Directional Coupler (DC)	0.27 dB	Lv D, Wu L, Liu C, Li A, Wang R, Wu A. Broadband and low-loss silicon photonic directional coupler for signal power tapping on the 3 μm soi waveguide platform. InPhotonics 2023 Jul 4 (Vol. 10, No. 7, p. 776). MDPI.
Propagation in waveguides	0.7 dB/cm	AIM Photonics, “Base passive photonic integrated circuit (PIC) process,” (2024).
Phase Shifters	0.23 dB	Ding F, Dai J, Chen Y, Zhu J, Jin Y, Bozhevolnyi SI. Broadband near-infrared metamaterial absorbers utilizing highly lossy metals. Scientific reports. 2016 Dec 21;6(1):39445.
Bend	0.086 dB (Bending Radius = 1μm)	Vlasov YA, McNab SJ. Losses in single-mode silicon-on-insulator strip waveguides and bends. Optics express. 2004 Apr 19;12(8):1622-31.
Taper	0.016 dB	Shen H, Fan L, Wang J, Wirth JC, Qi M. A taper to reduce the straight-to-bend transition loss in compact silicon waveguides. IEEE Photonics Technology Letters. 2010 May 24;22(15):1174-6.
Waveguide Crossing	0.03 dB	Lu W, Shi H, Ma C, Li Z, Zhu S, Wu Y, Tai R. Ultra-low loss SOI waveguide crossings designed by a hybrid global optimization based on deep learning. Optics Communications. 2024 Apr 1;556:130189.